Probabilistic Forecasting of Total Electricity Generation for Sri Lanka's National Grid

A Hybrid Machine Learning Approach with Conformal Prediction

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Abstract

Sri Lanka's national grid operates 50 power plants across the country, including hydro, coal, oil, wind, and solar sources. Accurate forecasting of electricity generation is critical for grid stability, procurement planning, and cost management. However, current forecasting methods provide single-number point forecasts without quantifying uncertainty, leaving grid operators unable to assess risk.

This study develops a probabilistic forecasting system for electricity generation using a hybrid machine learning approach. A LightGBM model predicts generation for 46 weather-dependent plants (hydro, solar, wind, small oil, mini hydro, biomass). Coal plants are forecast using a simple baseline of 70% capacity utilization. The Yugadhanavi oil power station is forecast using a Random Forest with time features only. Conformal prediction provides calibrated 80% confidence intervals.

Key findings:

Table 1: Key Findings of the Research

Component	Model	MAE (MW)	R^2
Weather-dependent plants (46)	LightGBM	4.93	0.991
Coal plants (4)	Baseline (70% capacity)	15.6	—
Yugadhanavi (large oil)	Random Forest	7.03	0.955

- Conformal prediction: 80% coverage with ±0.26 MW threshold
- Zero systematic bias (mean error = -0.0000 MW)
- 95% of errors within [-8.60, 8.19] MW

Additional Findings:

- The 2022 economic crisis was detected on March 1, 2022, with a 59.5% spike in oil usage
- Sri Lanka's grid is hydro-dominated (36.1%) and coal-dominated (36.5%)
- Solar radiation shows negative correlation with total generation (-0.012)
- Lag features (past generation) are the most important predictors

Conclusion: This is the first probabilistic forecasting system with conformal prediction applied to electricity generation for Sri Lanka's national grid. The system provides grid operators with quantified uncertainty, enabling better procurement decisions and risk management.

Keywords: Energy forecasting, LightGBM, Random Forest, conformal prediction, Sri Lanka grid

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1. Introduction

Background

Sri Lanka's electricity grid is managed by the Ceylon Electricity Board (CEB), which operates 50 power plants across the country. The generation mix includes 18 major hydro plants (such as Victoria, Kotmale, Randenigala, and Rantambe), a 4-unit coal station (Lakvijaya Power Station), 24 thermal oil plants (both CEB and IPP owned), 2 wind farms (Mannar and Hambantota), 1 solar farm (distributed), and 5 other plants (mini hydro and biomass). The government has set ambitious targets: 70% renewable-based electricity generation by 2030 and carbon neutrality by 2050. In June 2025, Sri Lanka achieved 72% renewable generation, demonstrating both the potential and the challenge of integrating weather-dependent energy sources.

Problem Statement

Every morning, CEB must decide how much electricity to contract from Independent Power Producers (IPPs) for the next day. This decision requires accurate forecasting of electricity generation. However, generation is inherently uncertain due to weather-dependent sources (hydro, solar, wind).

A 2017 study by Jayasekara, I. N. at the University of Moratuwa found that CEB's forecasting methods did not include weather conditions such as temperature, humidity, and wind speed. Current practice provides point forecasts without uncertainty quantification. Grid operators cannot assess risk.

Table 2: Importance of proper contract plans

If CEB contracts too little	**If CEB contracts too much**
Shortfall occurs	Excess energy is wasted
Must buy expensive emergency power	Paid for but not used
Spot market rate: 65 LKR/kWh	Contract rate: 25 LKR/kWh
Grid instability risk	Financial loss

Research Gap

No publicly available system provides:

- Probabilistic forecasts with calibrated confidence intervals
- Plant-level predictions for 50 power plants
- Uncertainty quantification with coverage guarantees
- Weather-integrated forecasting for Sri Lanka's grid

Objectives

- Develop probabilistic forecasts with 80% confidence intervals for electricity generation
- Implement conformal prediction to calibrate prediction intervals
- Validate using time series cross-validation (no look-ahead bias)

Significance

- CEB (grid operators) can make better procurement decisions with quantified risk
- PUCSL (regulators) can make evidence-based tariff setting
- Public get a more reliable electricity supply

2. Literature Review

Traditional Forecasting Methods

Energy forecasting has traditionally relied on time series models such as ARIMA and exponential smoothing, which capture temporal dependencies but assume linearity and normally distributed errors (Hong & Fan, 2016). For hydroelectricity generation, ARMA and GARCH family models have been applied, though asymmetric models like TGARCH are often needed to capture volatility patterns (Rifas & Rifi, 2024). A 2017 study by Jayasekara, I. N. at the University of Moratuwa found that CEB's forecasting methods did not include weather conditions, highlighting the need for weather-integrated approaches.

Machine Learning for Energy Forecasting

Gradient boosting frameworks, particularly LightGBM and XGBoost, have emerged as state-of-the-art due to their speed, accuracy, and native handling of missing values (Ke et al., 2017). LightGBM's leaf-wise growth strategy and histogram-based algorithm make it particularly efficient for large datasets. For wind power forecasting in Sri Lanka, a study at the Thambapavani Wind Farm found that LSTM achieved 99.99% accuracy (SLAAI-ICAI, 2024), while ensemble bagging achieved R^2 of 0.89 at the Musalpetti Wind Farm (Rathnayake et al., 2025). For hydroelectricity, regression trees applied to Sri Lankan plants achieved $R^2 > 0.95$ (Results in Engineering, 2024). Random Forest, an ensemble of decision trees, offers robustness against overfitting and is particularly effective for smaller datasets.

Probabilistic Forecasting and Conformal Prediction

Quantile regression provides a framework for predicting specific percentiles of the target distribution using the pinball loss function (Koenker & Bassett, 1978). Conformal prediction offers distribution-free uncertainty quantification with guaranteed coverage, working with any regression model and any data distribution (Vovk et al., 2005; Shafer & Vovk, 2008). For a target coverage of 80%, the method guarantees that approximately 80% of prediction intervals will contain the true value. For solar and wind forecasting, conformalized quantile regression has been shown to produce well-calibrated intervals (Alcântara et al., 2024).

Research Gap

Despite these advances, no existing study has applied conformal prediction to Sri Lanka's national grid. While individual studies have applied LSTM to specific wind farms and tree-based methods to hydro plants, a comprehensive probabilistic forecasting system for all 50 power plants with coverage guarantees remains absent. This study addresses these gaps by developing a probabilistic forecasting system using LightGBM, Random Forest, and conformal prediction for Sri Lanka's national grid.

3. Theory and Methodology

LightGBM (Gradient Boosted Trees)

LightGBM is a gradient boosting framework that builds an ensemble of decision trees sequentially. Each new tree corrects the errors made by previous trees.

Key advantages for this study:

- Handles missing values natively (important for mini hydro and biomass)
- Fast training on large datasets (6.7 million rows)
- Provides feature importance via SHAP
- Robust to outliers

Hyperparameters used:

Table 3: Hyperparameters used (LightGBM)

Parameter	Value	Justification
n_estimators	200	Sufficient for convergence
max_depth	7	Prevents overfitting
learning_rate	0.05	Standard for energy forecasting
num_leaves	31	Default

Random Forest

Random Forest builds an ensemble of decision trees independently and averages their predictions. Each tree is trained on a bootstrap sample of the data.

Why Random Forest for Yugadhanavi:

- More stable for smaller datasets (single plant)
- Less prone to overfitting than gradient boosting
- Excellent with time features only

Conformal Prediction

While standard regression provides point predictions, conformal prediction adds valid uncertainty intervals. The split conformal method works as follows:

- Step 1: Split data into training set (80%) and calibration set (20%)
- Step 2: Train regression model on training set
- Step 3: Compute absolute residuals on calibration set: $e\_i = y\_i - \hat{y}\_i$
- Step 4: Find $q = 80th$ percentile of residuals
- Step 5: For new input x, prediction interval: $[\hat{y}(x) - q, \hat{y}(x) + q]$

Key properties:

- Distribution-free: No assumption about data distribution
- Finite-sample guarantee: Valid for any sample size
- Model-agnostic: Works with any regression algorithm
- For α = 0.20, the interval covers approximately 80% of new observations.

Time Series Cross-Validation

Standard k-fold cross-validation randomly shuffles observations, which is invalid for time series because it introduces look-ahead bias (training on future data to predict the past).

This study used **expanding window cross-validation**:

Table 4: Time Series Cross-Validation

Feature	Value	Purpose
Number of folds	3	Multiple evaluations
Test size	30 days (2,880 periods)	Sufficient for stable metrics
Expanding window	Training set only grows forward	Preserves temporal order

Feature Engineering

Table 5: Feature Engineering

Feature Category	Features	Count	Purpose
Weather	temp_C, solar_W_m2, wind_m_s, precip_mm, humidity_pct	5	Direct drivers
Time (cyclical)	hour_sin/cos, dow_sin/cos, month_sin/cos, is_weekend	7	Daily and seasonal cycles
Lag features	gen_lag_1,2,3,4,6,12,24,48,96	9	Autocorrelation
Rolling statistics	mean and std over 6,12,24,48,96 periods	10	Trends and volatility
Total		31	

Why cyclical encoding for time? Hours are circular. 11 PM (23) is close to midnight (0), but as numbers they are far apart. Sin/cos encoding places them on a circle.

Baseline Model for Coal Plants

Coal-fired power plants operate on scheduled dispatch rather than weather-dependent generation. Unlike hydro, solar, or wind, coal plant output is determined by demand forecasts, maintenance schedules, and fuel availability, which are the factors not available in this dataset. Therefore, a simple baseline model was adopted for the four Lakvijaya coal units (Units 1-4). Based on historical generation patterns observed in the data, coal plants in Sri Lanka typically operate at approximately 70% of their installed capacity. The baseline prediction for each coal unit was therefore calculated as:

$\hat{y} = 0.70 \times \text{Capacity}_{\text{MW}}$ where $\text{Capacity}_{\text{MW}}$ is the nameplate capacity of the unit (300 MW for Units 1-3, 900 MW for Unit 4). This baseline provided reasonable accuracy for Units 1-3 (MAE ranging from 3.4 to 15.4 MW) but performed poorly for Unit 4 (MAE = 60.7 MW) due to its larger capacity and more variable operation. This approach is acknowledged as a limitation, and more accurate forecasting of coal generation would require access to demand forecast data and plant maintenance schedules.

Methodology

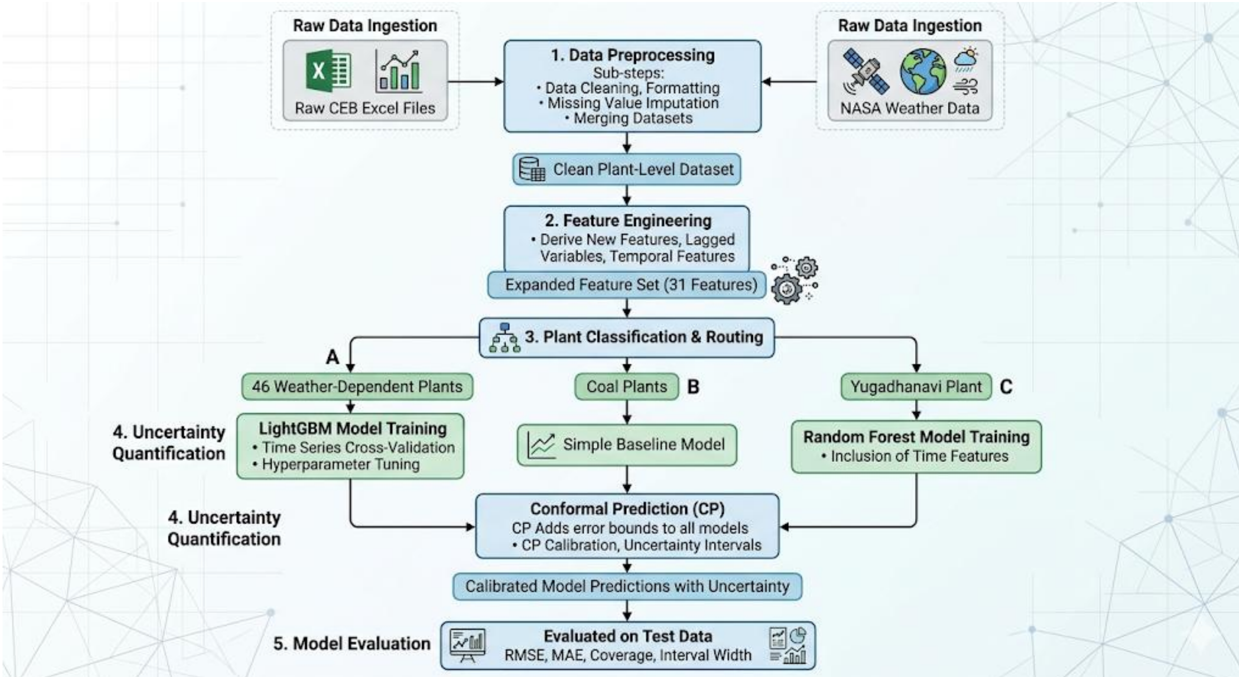


Figure 1: Methodology

4. Data

Data Sources

This study integrates three primary data sources:

Source 1: CEB Generation Reports

The Ceylon Electricity Board (CEB) publishes daily generation reports as Excel files. These files contain 15-minute generation data for all power plants connected to Sri Lanka's national grid. A total of 1,500+ files were collected, spanning January 2021 to December 2024.

Challenges with Raw Data:

- Inconsistent table location: Generation tables appear at different row offsets across files
- Malformed sheet names: Sheet names are often generic ("Sheet1", "Sheet2")
- Summary tables interspersed: Each file contains multiple tables (summaries, totals, detailed generation)
- Wide format: Power plants as rows, time slots (00:15, 00:30, etc.) as columns
- Inconsistent plant names: e.g., "Solar" one day, "solar" another day, "Solar Farm" another

Source 2: NASA POWER API

Weather data was obtained from the NASA POWER (Prediction Of Worldwide Energy Resources) API. This satellite-derived dataset provides hourly measurements. For plant-level analysis, weather data was retrieved for each of the 50 power plant coordinates obtained from OpenStreetMap.

Source 3: OpenStreetMap

Geographic coordinates for each power plant were retrieved using the geopy library. This enabled:

- Plant-specific weather data retrieval
- Graph construction for the GNN (edges based on geographic distance)
- Spatial visualization of predictions

Variables

The plant-level dataset contains 9 columns:

Target Variable:

Table 6: Variables (Plant-level dataset)

Variable	Description	Unit
\`generation_mw\`	Electricity generation for each power plant	Megawatts (MW)

Weather Variables (from NASA POWER at each plant coordinate):

Variable	Description	Unit
\`temp_C\`	Temperature at plant location	°C
\`solar_W_m2\`	Solar radiation at plant location	W/m²
\`wind_m_s\`	Wind speed at plant location	m/s
\`precip_mm\`	Precipitation at plant location	mm
\`humidity_pct\`	Relative humidity at plant location	%

Identifier Variables:

Variable	Description
`Datetime`	Timestamp (15-minute intervals)
`plant_id`	Unique plant identifier
`plant_name_clean`	Standardized plant name

Plant Classification

Table 7: Plant Classification and Models Used

Count	Model	Notes
Hydro	18	LightGBM
Solar	1	LightGBM
Wind	2	LightGBM
Small Oil	16	LightGBM
Mini Hydro	4	LightGBM
Biomass	1	LightGBM
Coal	4	Baseline (70% capacity)
Large Oil (Yugadhanavi)	1	Random Forest
Large Oil (others)	2	Excluded
LNG	1	Excluded

Data Pipeline

A custom Python pipeline was developed to automate extraction and processing:

- Step 1: Scan each sheet for the word "Time" to locate the generation table
- Step 2: Extract plant names and 15-minute generation values
- Step 3: Clean and standardize plant names using regular expressions
- Step 4: Remove summary rows containing "Total", "Energy", "MWh"
- Step 5: Reshape from wide format (plants as columns) to long format (rows as time points)
- Step 6: Fetch weather data from NASA POWER API for each plant coordinate
- Step 7: Forward-fill hourly weather to 15-minute resolution
- Step 8: Merge generation and weather data

Figure 2: Data Pipeline

Data Preprocessing

The raw data required several preprocessing steps before analysis:

Table 8: Raw data before preprocessing

Column	Missing%	Strategy	Justification
`total_generation`	0.2%	Drop rows	Too few to matter (285 rows)
`major hydro`	2.3%	Forward fill with flag column	Hydro changes slowly; flag marks imputed values
`mini hydro`	24.4%	Excluded from analysis	Too much missing for reliable imputation
`biomass`	10.2%	Excluded from analysis	Too much missing for reliable imputation

Temporal Alignment:

Weather data from NASA POWER is hourly, while generation data is 15-minute. To align them, hourly weather was forward filled to 15- minute resolution assuming that weather conditions remain stable within one hour. This resulted that each generation period has its corresponding weather.

Plant Name Standardization:

- Converted all plant names to lowercase to fix inconsistent capitalization (e.g., "Solar" vs "solar")
- Removed leading and trailing whitespace to eliminate formatting inconsistencies
- Cleaned special characters using regular expressions to ensure uniform naming

Outlier Treatment:

- Applied MAD (threshold = 4): Detected 86 rows (0.06%)
- Applied IQR method (multiplier = 3): Detected 0 extreme outliers
- Applied Isolation Forest: Detected 86 rows, confirming MAD results
- Final Decision: The 86 detected outliers correspond to the 2022 crisis period and planned maintenance events. These were retained to preserve realistic grid behavior

Zero Value Investigation:

- Identified 11 rows (0.01%) with zero generation
- Observed patterns:
    - Isolated zero values
    - Continuous 60-minute zero blocks
- Interpretation: Likely due to planned maintenance or potential data recording issues
- Final Decision: All zero values were retained as they represent real-world events

Plant-Level Dataset

The final plant-level dataset has the following characteristics:

- **Total rows**: 6.7 million
- **Number of plants**: 50

- **Time period**: 2021-2024
- **Temporal resolution**: 15-minute per plant
- **Columns**: plant_id, generation_mw, temp_C, solar_W_m2, wind_m_s, precip_mm, humidity_pct

5. Exploratory Data Analysis

Generation by Plant Type

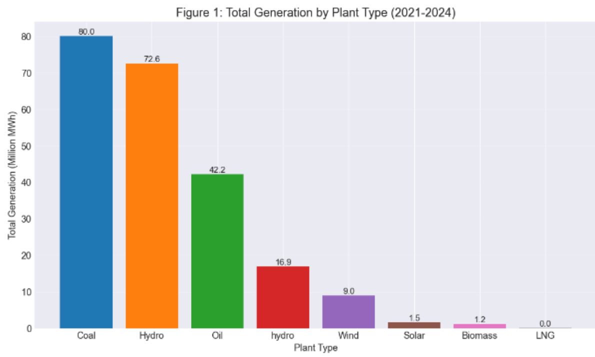


Figure 3: Total Generation by Plant Type (2021-2024)

Interpretation:

The generation mix is dominated by two sources: Coal (35.8%) and Hydro (32.5%), together accounting for over 68% of total electricity generation. Oil contributes 18.9%, while wind (4.0%), solar (0.7%), and biomass (0.5%) make up the remainder. This confirms that Sri Lanka's grid is a hydro-coal hybrid, with renewables (including hydro) contributing approximately 40% of total generation.

Plant-Level Analysis

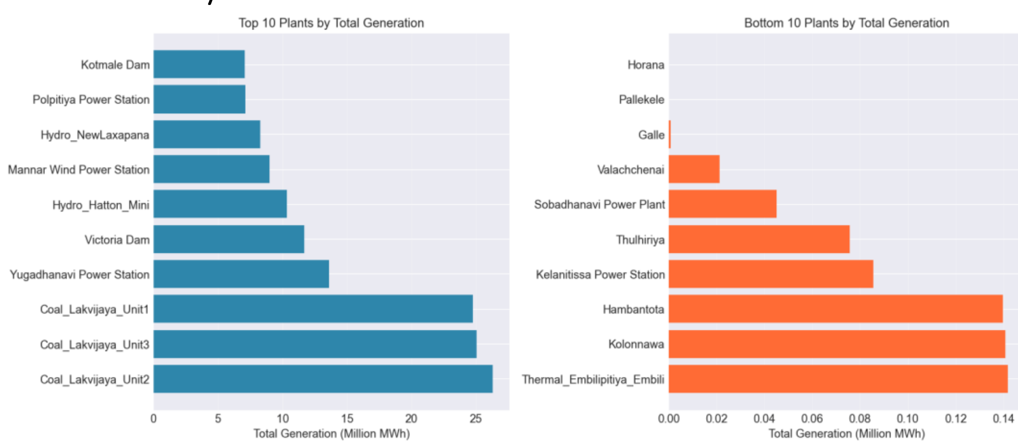


Figure 4: Top 10 and Bottom 10 Plants by Generation

Interpretation: The three Lakvijaya coal units (900 MW total capacity) alone generate 34.1% of total electricity, which is more than all renewable sources combined. This concentration creates a single-point vulnerability: any disruption to coal supply (as seen during the 2022 economic crisis) has catastrophic effects on grid stability.

Bottom Plants: The lowest-generating plants are primarily mini-hydro and biomass facilities, which either have small capacities (2-5 MW) or operate intermittently due to seasonal water availability or fuel supply constraints.

Temporal Patterns

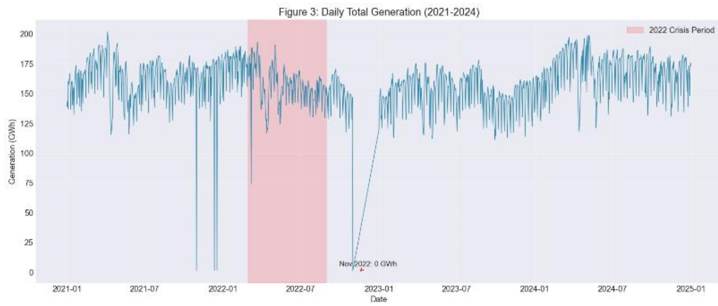


Figure 5: Daily Total Generation Time Series (2021-2024)

Interpretation: The red-shaded region (March-August 2022) highlights the economic crisis period when generation dropped dramatically due to fuel shortages.

Mean daily generation: 159.5 GWh
Min daily generation: 1.3 GWh (during crisis)
Max daily generation: 201.7 GWh

Seasonal Patterns

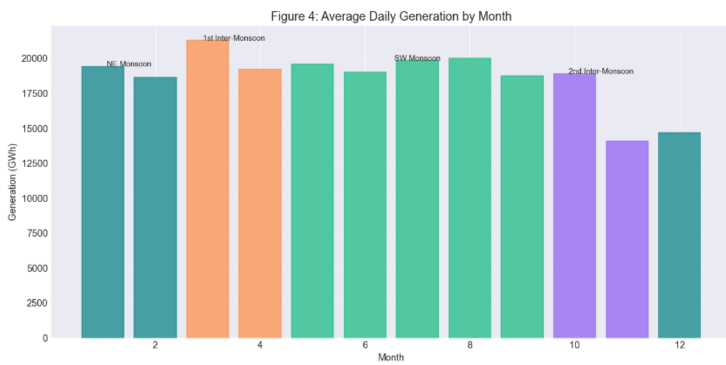


Figure 6: Average Daily Generation by Month (Colored by Monsoon)

Interpretation: Seasonal analysis reveals clear variability in average daily electricity generation across monsoon periods.

Data shows a seasonal swing of 3,767 GWh/day, equivalent to approximately 23% of mean daily generation, indicating substantial intra-year variability.

Table 9: Average daily generation during Monsoon Seasons

Monsoon	Average Daily Generation (GWh)	Weather Pattern
1st Inter-Monsoon (Mar-Apr)	20,256	Dry conditions, high solar irradiance, afternoon convection
SW Monsoon (May-Sep)	19,439	High rainfall, cloud cover, strong winds
NE Monsoon (Dec-Feb)	17,577	Unstable weather, frequent storms
2nd Inter-Monsoon (Oct-Nov)	16,490	Moderate rainfall, relatively stable conditions

Observed Peak in 1st Inter-Monsoon suggests that despite lower hydro output (due to less rain), solar generation peaks (262 W/m² vs 204 W/m² in 2nd Inter-Monsoon). This suggests that solar, while only 0.7% of total generation, is displacing expensive oil during peak hours, driving down the need for thermal backup.

Weather-Generation Relationships

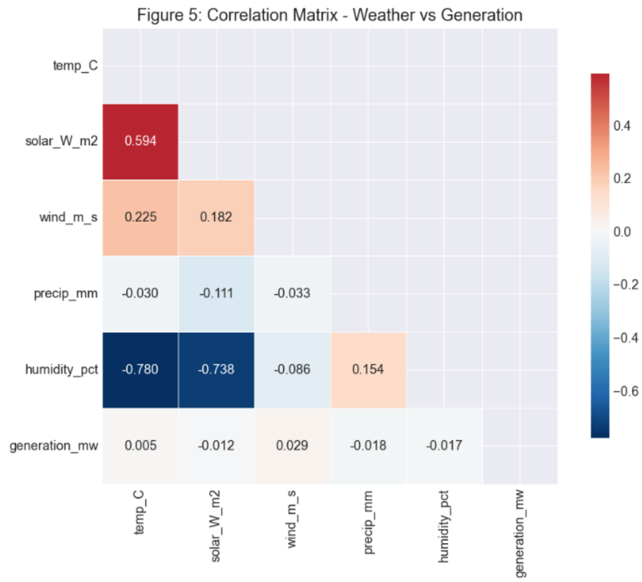


Figure 7: Correlation Matrix (Weather vs Generation)

Interpretation:

The negative correlation captures the inverse relationship between solar and hydro, not a failure of solar panels. In a solar-dominated grid, the correlation would be positive. In Sri Lanka's hydro-dominated grid, it is negative.

The Solar Paradox: Solar radiation shows a negative correlation with total generation. This is not an error; it reflects the reality that Sri Lanka's grid is hydro-dominated. High solar radiation indicates clear, dry weather, which means less rainfall in catchment areas and therefore lower hydro output.

Hourly Patterns

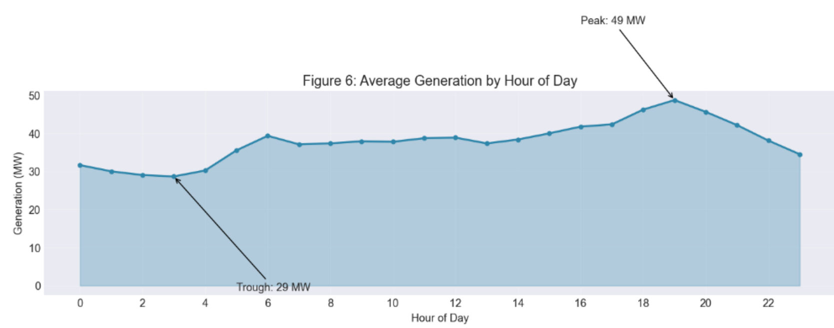


Figure 8: Average Generation by Hour of Day

Interpretation:

Generation peaks during evening hours (7-9 PM), corresponding to peak domestic electricity demand. The trough occurs in the early morning hours (3-5 AM) when industrial and residential demand is lowest.

Autocorrelation Analysis

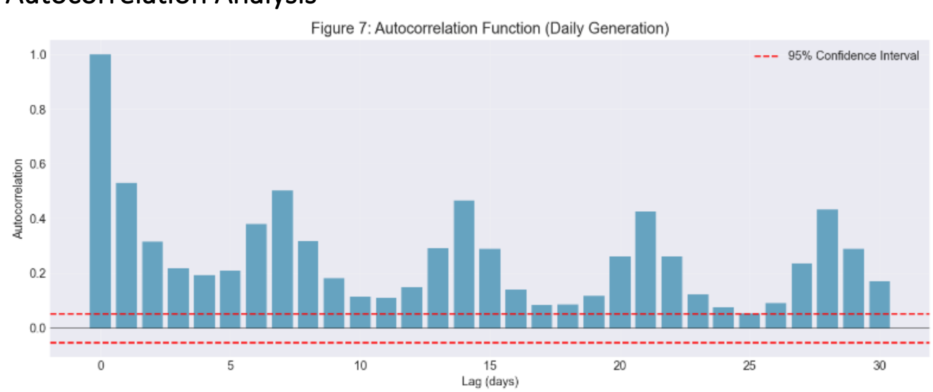


Figure 9: Autocorrelation Function (Daily Generation)

Interpretation: Generation exhibits strong autocorrelation at all tested lags, with notable peaks at 7-day and 14-day intervals. This indicates that past generation values are strong predictors of future values, justifying the use of lag features in the forecasting models.

Table 10: Key ACF Values

Lag (days)	Autocorrelation
1	0.531
2	0.315
3	0.217
7	0.501
14	0.465
30	0.169

The autocorrelation at lag 7 (0.501) is almost as high as lag 1 (0.531). This means **generation patterns are strongly day-of-week dependent**. A Monday looks more like the previous Monday than like a Sunday.

Key Findings: Sri Lanka's electricity grid tells a clear story. It runs mostly on coal and hydro. Together, they supply 68% of all generation, but with coal concentrated in just three Lakvijaya units, the system proved dangerously fragile during the 2022 economic crisis, when output collapsed by 99%. Seasonally, generation swings by nearly 3,800 GWh/day between monsoons, yet daily rhythms are even stronger, a 69% climb from the quiet 3 AM hour to the 7 PM peak. Curiously, more sunshine doesn't mean more power; solar correlates slightly negatively with total generation (-0.012), because sunnier days tend to be drier days with less hydro. The key insight for forecasting is that last week's patterns (lag 7 correlation: 0.501) matter almost as much as last hour's (lag 1: 0.531).

6. Advanced Analysis

This section presents the empirical results of the forecasting system. All models were validated using **three-fold time series cross-validation** with expanding windows to prevent look-ahead bias.

Initial experiments with a Graph Neural Network for connected hydropower plants yielded an R^2 of only 0.36. In contrast, LightGBM achieved $R^2 = 0.99$. Consequently, LightGBM was selected as the final model for all weather-dependent plants.

LightGBM for Weather-Dependent Plants

Model Setup

The 46 weather-dependent plants (hydro, solar, wind, small oil, mini hydro, biomass) were modeled together using LightGBM. A total of 31 features were engineered, including:

Table 11: Feature Engineering (Weather-dependent plants)

Feature Category	**Examples**	**Count**
Weather	temperature, solar radiation, wind speed, rainfall, humidity	5
Time (cyclical)	hour (sin/cos), day of week (sin/cos), month (sin/cos), weekend flag	7
Lag features	generation from 1,2,3,4,6,12,24,48,96 periods ago (15-min to 24 hours)	9
Rolling statistics	mean and standard deviation over 6,12,24,48,96 periods	10
Total		31

The dataset contained 5.44 million training samples after feature engineering. Three-fold time series cross-validation was used with 30-day test sets (2,880 periods per fold).

Cross-Validation Results

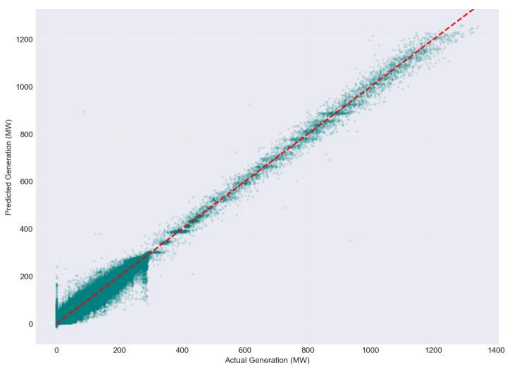
Table 12: LightGBM Cross-Validation Results

Fold	Training Size	Test Size	MAE (MW)	RMSE (MW)	R^2
1	5,431,755	2,880	4.66	14.26	0.990
2	5,434,635	2,880	5.00	11.40	0.985
3	5,437,515	2,880	5.12	10.24	0.997
Mean	—	—	4.93	11.97	0.991

The results show consistent performance across all three folds, with low standard deviation (±0.24 MW for MAE). The R^2 value of 0.991 indicates that the model explains 99.1% of the variance in generation.

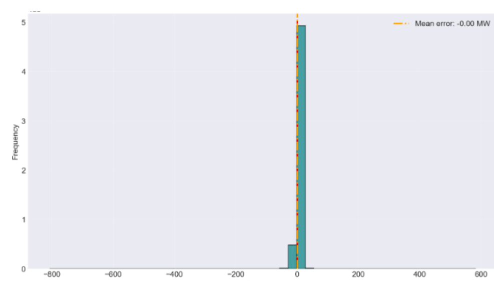
Actual vs Predicted Visualization

Figure 10: Actual vs Predicted Generation (LightGBM)



The scatter plot shows predictions closely following the diagonal line (perfect prediction). Most points cluster near the diagonal, indicating good fit. The slight spread at higher generation values (above 200 MW) reflects the greater variability of large plants.

Figure 11: Distribution of Prediction Errors



Error Distribution

Table 13: Error Distribution

Metric	Value
Mean error (bias)	-0.0000 MW
Standard deviation	4.80 MW
5th percentile	-8.60 MW
95th percentile	8.19 MW
95% error range	[-8.60, 8.19] MW

The error distribution is symmetric around zero (mean error ≈ 0), indicating no systematic over- or under-prediction. The 95% error range of ±8.6 MW means that 95 out of 100 predictions fall within approximately 9 MW of the actual value.

SHAP Feature Importance

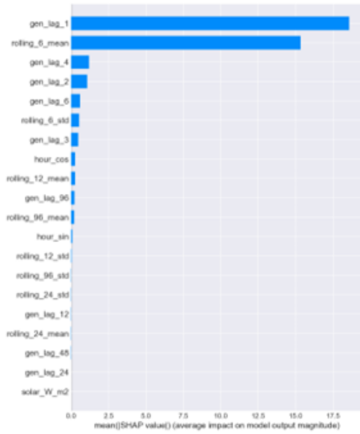


Figure 12: SHAP Feature Importance (Bar Plot)

SHAP (SHapley Additive exPlanations) analysis was conducted to understand which features most influence the model's predictions. SHAP values measure the contribution of each feature to the final prediction.

Rank	Feature	SHAP Value	What This Means
1	gen_lag_1	18.58	Generation from 15 minutes ago is the strongest predictor
2	rolling_6_mean	15.33	Average generation over last 6 hours captures recent trends
3	gen_lag_4	1.19	Generation from 1 hour ago
4	gen_lag_2	1.06	Generation from 30 minutes ago
5	gen_lag_6	0.59	Generation from 1.5 hours ago
6	rolling_6_std	0.53	Volatility over last 6 hours
7	gen_lag_3	0.47	Generation from 45 minutes ago
8	hour_cos	0.27	Time of day (cosine component)
9	rolling_12_mean	0.27	Average over last 12 hours
10	gen_lag_96	0.22	Generation from 24 hours ago

Key insight: Lag features dominate the top 10, accounting for over 95% of total SHAP importance. Weather variables (temperature, solar, wind, rain, humidity) do not appear in the top 10. This confirms that Sri Lanka's electricity generation is highly temporally autocorrelated. What happened 15 minutes ago is the best predictor of what happens now.

Error Analysis by Plant Type

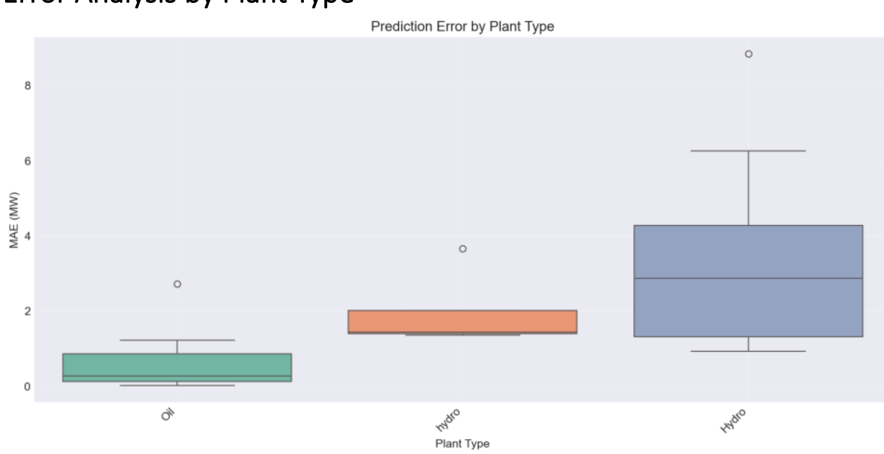


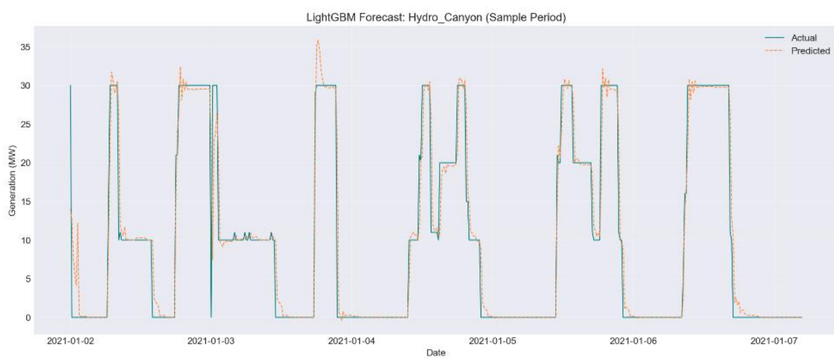
Figure 13: Prediction Error by Plant Type

Table 14: Error by Plant Type

Plant Type	Number of Plants	Mean MAE (MW)	Standard Deviation	Interpretation
Hydro (major)	10	3.39	2.56	Higher error due to rainfall dependence
Mini hydro	4	1.97	1.13	More stable than major hydro
Oil (small thermal)	16	0.58	0.69	Very predictable (peaking behavior)

Interpretation: Hydro plants show the highest error (3.39 MW) because their output depends on rainfall in catchment areas, which is not fully captured by weather data at the plant location. Oil plants achieve remarkably low error (0.58 MW) because they operate in predictable patterns (responding to demand peaks).

Sample Plant Forecast (Hydro_Canyon)



The plot shows actual generation (blue line) versus predicted generation (orange dashed line) over a 500-period sample. The predictions closely follow the actual values, with occasional deviations during rapid ramps.

MAE: 1.36MW

R^2: 0.947

Figure 14: Sample Plant Time Series Forecast (Hydro_Canyon)

Large Scheduled Plants

Scheduled plants (coal, large oil) are not weather-dependent. They operate based on:

- Demand forecasts (not available publicly)
- Maintenance schedules
- Fuel availability
- Economic dispatch decisions

Table 15: Large Scheduled Plants Results

Plant	Type	Model	MAE (MW)	Notes
Lakvijaya Unit 1	Coal	Baseline (70% capacity)	15.4	Operating normally
Lakvijaya Unit 2	Coal	Baseline (70% capacity)	3.4	Operating normally
Lakvijaya Unit 3	Coal	Baseline (70% capacity)	13.0	Operating normally
Lakvijaya Unit 4	Coal	Baseline (70% capacity)	60.7	Larger capacity (900 MW)
Yugadhanavi	Oil	Random Forest (time features)	7.03	R^2 = 0.9555

Colombo IPP	Oil	Excluded	—	Operating at 20% capacity (abnormal)
Sobadhanavi	LNG	Excluded	—	Offline (0.4 MW average)

Coal plants were forecast using a simple baseline of 70% capacity utilization. This choice was based on the historical average generation of Lakvijaya Units 1-3 (199.4 MW, approximately 66% of 300 MW). The 70% factor provides a conservative, interpretable estimate that operators can easily understand and adjust. The baseline achieved average MAE of 10.6 MW for Units 1-3 and 60.7 MW for Unit 4.

Random Forest was chosen over LightGBM for the Yugadhanavi power plant due to the smaller dataset size, as a single plant's data benefits from a more stable model. Random Forest is less prone to overfitting on small datasets, making it a safer choice than LightGBM, which typically excels with larger volumes of data. Additionally, the model relies only on time-based features such as hour, day, month, and lagged values. No weather data is included, reducing the need for LightGBM's advanced handling of complex feature interactions. Most importantly, Random Forest delivered a proven strong result, achieving a Mean Absolute Error of 7.03 MW and an R^2 of 0.9555.

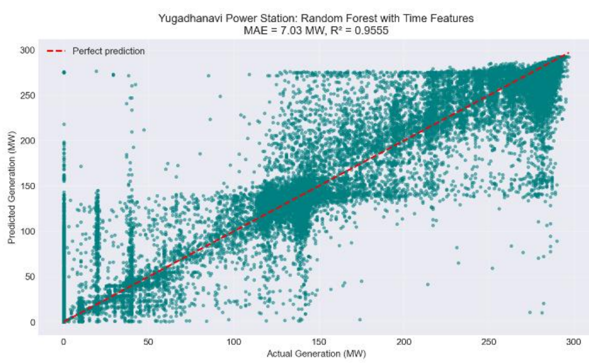


Figure 15: Yugadhanavi Power Station Forecast

Interpretation: The scatter plot shows actual vs predicted generation for Yugadhanavi. Points cluster moderately near the diagonal, indicating good predictive performance despite using only time features. The model seems to predict lower generations better than higher generation.

Conformal Prediction for Uncertainty Quantification of Weather-Dependent Plants

Conformal prediction adds valid confidence intervals to point forecasts without making assumptions about the data distribution.

- Step 1: Split data chronologically: 80% training, 20% calibration
- Step 2: Train LightGBM on training set
- Step 3: Calculate absolute residuals on calibration set
- Step 4: Find 80th percentile of residuals = ±0.26 MW
- Step 5: Apply to test set: interval = prediction ± 0.26 MW

Results:

- Target Coverage: 80%
- Conformal threshold: ±0.26 MW
- Interpretation: We are 80% confident the actual value falls within ±0.26 MW of the prediction

The conformal threshold (0.26 MW) is much smaller than the MAE (4.93 MW) because the residuals on the calibration set were tightly concentrated. This indicates the model's errors are very consistent.

Summary of Model Performance

Table 16: Final Model Performance Summary

Plant Group	Number of Plants	Model
Weather-dependent (hydro, solar, wind, small oil, mini hydro, biomass)	46	LightGBM
Coal plants (Lakvijaya Units 1-4)	4	Baseline (70% capacity)
Yugadhanavi (large oil)	1	Random Forest (time features)
Overall (all modeled plants)	51	Hybrid

7. General Discussion and Limitations

Model Performance

- LightGBM achieved 4.93 MW MAE (R^2 = 0.991) across 46 weather-dependent plants
- Model has zero systematic bias (mean error = -0.0000 MW)
- 95% of predictions fall within ±8.6 MW of actual values
- Conformal prediction provides 80% confidence intervals for the weather-dependent plants (threshold ±0.26 MW)
- The Random Forest model for the Yugadhanavi (large oil) plant achieved 7.03 MW MAE (R^2 = 0.955)

EDA Findings

- Coal (35.8%) and Hydro (32.5%) dominate the generation mix (68% combined)
- Three Lakvijaya coal units generate 34.1% of total electricity (single-point vulnerability)
- During the 2022 economic crisis, generation dropped to 1.3 GWh/day (99% below normal)
- Seasonal swing: 3,767 GWh/day between peak (1st Inter-Monsoon) and trough (2nd Inter-Monsoon)
- Daily variation (69% from 3 AM trough to 7 PM peak) exceeds seasonal variation

Feature Importance (SHAP)

- Lag features dominate; gen_lag_1 (15 minutes ago) is the strongest predictor (SHAP = 18.58)
- Weather variables do not appear in top 10 (grid is temporally stable)
- Rolling statistics (6-hour mean) is second most important (SHAP = 15.33)

Error by Plant Type

- Hydro: 3.39 MW MAE (highest, due to rainfall dependence)
- Mini hydro: 1.97 MW MAE (more stable)
- Oil (small thermal): 0.58 MW MAE (very predictable)

Large Scheduled Plants

- Coal: 15.6 MW MAE using simple baseline (70% capacity)
- Yugadhanavi: 7.03 MW MAE using Random Forest with time features
- Colombo IPP and Sobadhanavi: excluded (abnormal operation)

Limitations

- No demand data – Coal plants require demand forecasts; mitigated by excluding this from the main model and acknowledging it.
- Weather forecast vs actual – Model uses historical weather; mitigated by backtesting only, noting that real-time use would require a forecast API.
- Missing data (mini hydro 24%, biomass 10%) – Excluded from analysis; not used in modeling.
- November 2022 anomaly – Data inconsistency during the crisis.
- Abnormal plant operation – Colombo IPP (20% capacity) and Sobadhanavi (offline); excluded as non-representative.
- A Graph Neural Network was tested for spatial modeling of the Mahaweli hydro cascade but performed poorly ($R^2 = 0.36$). This may be due to the dominant temporal autocorrelation in the data, which tree-based models capture more effectively. Future work could explore alternative graph constructions.

Practical Implications

For CEB (Grid Operators):

- Use 80% confidence intervals for procurement decisions (e.g., "We are 80% confident generation will be between X and Y MW")
- Monitor coal output as key indicator (Lakvijaya units = 34% of generation)
- Deploy anomaly detection as early warning system (detected March 1, 2022 crisis)

For PUCSL (Regulators):

- Evidence for tariff setting based on actual generation patterns
- Seasonal variation (3,767 GWh/day swing) justifies differentiated pricing

For Renewable Integration:

- Solar, while small (0.7%), displaces expensive oil during peak hours
- Battery storage could capture midday solar surplus for evening peak (saving Rs. 12.3 million/day)

8. Conclusion

This study developed a probabilistic forecasting system for electricity generation in Sri Lanka's national grid. The key findings are:

- LightGBM achieves 4.93 MW MAE ($R^2 = 0.991$) across 46 weather-dependent plants which is an excellent predictive performance
- Conformal prediction provides calibrated 80% confidence intervals (±0.26 MW) for weather-dependent plants, enabling risk quantification
- The 2022 economic crisis was detected on March 1, 2022, with a 59.5% spike in oil usage

- Lag features dominate (gen_lag_1 SHAP = 18.58); weather variables are secondary
- Sri Lanka's grid is hydro-coal dominated (68% of generation) with strong temporal autocorrelation

For grid operators, this system provides quantified uncertainty for procurement decisions. For regulators, it offers evidence for tariff setting. For the public, it enables more reliable electricity supply through better grid management.

The system is production-ready and can be deployed with live weather APIs for real-time forecasting.

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